Dikshant Agrawal

M.Sc. Embedded Systems and Microelectronics (EE&IT)
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ABOUT ME

Embedded engineer pursuing M.Sc. in Electrical Engineering and Information Technology (Major: Embedded Systems and Microelectronics) at Hochschule Darmstadt, with strong academic foundation and over 3 years of industry experience. I write RTOS and bare-metal firmware in C/C++ on ARM Cortex-M microcontrollers, down to register-level, interrupt handling, and JTAG/SWD bring-up, alongside hardware design in KiCad. I actively use Claude Code as a coding agent for complex device-driver development based on my architectural design, test-case generation and debugging workflows bringing AI-accelerated development into real embedded projects. Seeking a six-month full-time internship position to own and explore the intersection of hardware and firmware projects, aiming to develop innovative solutions through next-generation AI-assisted EE workflows.

WORK EXPERIENCE

Research Student - Firmware Development *Mar 2026 – Present*
Hochschule Darmstadt University of Applied Sciences, Darmstadt (Germany)

- Extending the CDD and Zigbee RF stack of a PSoC5 remote of student-cars from 1:1 to 1:n communication, extending the hierarchical state machine while keeping full backward compatibility.
- Writing and executing low-level drivers and unit tests, integrated with the car team through CI/CD workflow.
- Using Claude Code for more flexibility in firmware iteration and testing.

Embedded Systems Engineer *Jan 2023 – Aug 2025*
IITI DRISHTI CPS Foundation - Translational Research Park, IIT Indore (India)

- Built real-time, interrupt-driven firmware for human-body vitals monitoring using an Espressif dual-core MCU; the device was validated at three clinical sites.
- Wrote object-oriented C drivers for a BLE sampling and real-time signal processing on a Nordic nRF52840 wearable for athlete performance analysis.
- Built a GPS-GSM-enabled IoT weighing scale connecting over MQTT, enabling automated fleet and waste tracking for local governing bodies.
- Developed Python tooling for sensor calibration and signal validation.

Research and Development Engineer *Sep 2022 – Jan 2023*
Scientech Technologies Pvt. Ltd. (India)

- Designed embedded training kits covering IoT and RF links (Zigbee) communication for wireless energy monitoring, with hardware validation automated in Python.

Hardware Design Intern *Jan 2022 – Jul 2022*
Einfochips Pvt. Ltd. – an Arrow Company (India)

- Designed a buck converter using OrCAD for an aerospace project, covering power budget planning, component selection, symbol creation, schematic capture, and PCB layout.
- Verified I2C, SPI, UART and USB 2.0 interfaces during board bring-up; used JTAG for firmware verification and fault detection.

PROJECTS

Electronic Gas Pedal *Mar 2026 – Present*

- A closed-loop PID-controlled engine with gas pedal using AUTOSAR RTE (light version) where runnables communicate only via generated signal objects.

4th order band pass filter with CFAR *Apr 2026 – June 2026*

- Bare-metal firmware with schematic capture and a four-layer PCB for a cascaded OPA340 bandpass filter; analyzed the response in MATLAB (FFT/DFT, windowing for spectral leakage) with attention to signal integrity.

STM32N6 Edge-AI *Apr 2026 – Present*

- Trained a multi-head MLP in PyTorch to drive a Doom agent, exported via ONNX to C headers using STM32Cube AI Studio, and ran inference on the STM32N6 Neural-ART NPU.

EDUCATION

Master of Science in Electrical Engineering and Information Technology *Sep 2025 – Present*
Major: Embedded Systems and Microelectronics
Hochschule Darmstadt University of Applied Sciences, Darmstadt (Germany)
Coursework: Embedded Architecture & OS, Hardware Design, Signal Processing, FPGA, VLSI.

Bachelor of Technology in Mechatronics *Jul 2018 – Aug 2022*
Shri Vaishnav Institute of Technology and Science, SVVV, Indore (India)
Thesis: IoT-based baby monitor on ESP32 with real-time sensor analysis and alert system.

High School Diploma – Science and Mathematics *Apr 2017 – Apr 2018*
St. Arnold's Higher Secondary School, CBSE, Indore (India)

SKILLS

Embedded: RTOS (FreeRTOS and Erika OSEK), Bare-metal, CDD, register-level drivers, DMA, ARM Cortex-M, Intel MAX 10 (DECA), IoT, schematic capture, PCB design, analog and digital circuit design, Hardware-in-the-loop.
Communication Protocols: UART, I2C, SPI, USB 2.0, CAN
Programming Languages: C, C++, Python, MySQL
Instruments: Logic analyzer, PicoScope, oscilloscope, multimeter, DC Supply, Soldering
Tools: Claude Code, Git, Gitlab, KiCad, MATLAB, Vivado, Fusion 360, AWS
Soft Skills: Scientific writing, Presentation, Collaboration
Language: English: Fluent (C1), German: Beginner (A2), Hindi: Native

CERTIFICATIONS

PCB Designing and Material Training, Hochschule Darmstadt *Apr 2026*
Introduction to Embedded Systems and IoT by University of California, Irvine – Coursera *Mar 2025*
Embedded System Engineer by Udemy *Nov 2024*
Python Programming by IIT Indore *Feb 2023*

Date: 03.06.2026
Darmstadt, Germany